

# DESTER: Problems and Solutions

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July 2026

*A record of the substantive obstacles and decisions in the research build*

## PURPOSE

This document records the substantive obstacles, design decisions, and honest research findings encountered during DESTER’s build. Tactical noise — paste errors, terminal quirks, CSS regressions — is excluded. Every entry below either changed a design decision, revealed a research finding, or would meaningfully affect a future reader’s understanding of the project.

## DATA FOUNDATION

### DB1B discontinuation and the domestic pivot

The initial project scope contemplated an international alliance joint-venture case study (for example, Munich–Nashville under the Lufthansa/United A++ JV). DB1B, however, is domestic-only — the Bureau of Transportation Statistics does not release international O&D fares in this dataset. Additionally, DB1B was formally discontinued in July 2025 and replaced with DB1C (monthly, 40% sample). The project pivoted to a domestic connecting market (AUS-SLC on Delta) with the 2025 Q2 quarter — the final quarter of DB1B before the DB1C switchover. The international variant is documented as a future direction contingent on international O&D data availability.

### Year-swap during initial development

DB1BMarket was initially downloaded for 2024 Q2 as a development sample. Once the pipeline was stable, the year was swapped to 2025 Q2 to align with the operational context of the running market’s route decision. Because Layer 0 was correctly built with globbed CSV paths rather than hardcoded filenames, no code changes were required.

## THE SHARE MODEL’S TRANSFERABILITY FAILURE

The most substantive research finding of the project.

### Symptom

Wave 1’s multinomial logit share model, calibrated on 219 markets and backtested at 13.34pp MAE, failed catastrophically when applied to the second rigorously-analyzed market (ATL-SAT). Delta’s predicted share on ATL-SAT under the unmodified model was 0.9%, against observed share of 58.5% — a  $64\times$  miscalibration.

### Diagnosis

Ultra-low-cost carriers Frontier (F9) and Spirit (NK) had large positive fixed-effect intercepts fit to a subset of training markets where they held unusually high share. On AUS-SLC, F9 and NK have small local share (roughly 3% and 2%), so the inflated intercepts had little effect. On ATL-SAT, both ULCCs have real presence (F9 about 2%, NK about 19%), and the inflated intercepts dominated the softmax denominator — predicting F9 at 80% share and squeezing Delta down to near-zero.

### Two remedies that failed cleanly

*Iteration 2: minimum row-count filter.* Drop carriers with fewer than 200 training rows. F9 (approximately 800 rows) and NK (approximately 700 rows) sailed through. MAE regressed to 16.94pp because 16 legitimate small carriers (Endeavor, Alaska, JetBlue, Allegiant, and others) were dropped without touching the ULCCs.

*Iteration 3: minimum-distinct-markets filter.* F9 appears in 139 distinct training markets, NK in 94 — both compa-

rable to major incumbents (AA=142, UA=144). Neither was dropped. MAE stayed flat.

Both iterations stopped at their pre-registered stop conditions.

### The fix that worked

Iteration 4 applied post-fit empirical-Bayes-style shrinkage on carrier intercepts:

$$\beta_c^{\text{new}} = \beta_c^{\text{old}} \cdot \frac{1}{1 + \lambda \cdot \sigma_c^2}$$

where  $\sigma_c^2$  is the variance of carrier  $c$ 's per-market share-prediction residual across the training set. This targets the actual failure mode: high-variance-of-fit intercepts (unstable fits reflecting niche high-share markets) get shrunk toward zero, while low-variance intercepts (stable estimates on broadly-supported carriers) pass through nearly unchanged.

At  $\lambda = 10$ , F9 was shrunk 51.1% (cleared 40% threshold), NK was shrunk 39.1% (missed by 0.9pp). MAE improved to 10.28pp but the pre-registered stop condition triggered.

At  $\lambda = 15$  (Iteration 5, chosen from candidates 15 and 20 via the lower- $\lambda$  tiebreak), F9 shrunk 61.0%, NK shrunk 49.0%. MAE improved to 9.94pp — the best of the project, clearing not just the 15pp danger threshold but the 10pp publishable-quality target. Applied to ATL-SAT, Delta's predicted share rose from 0.9% to 23.8% — a  $26\times$  improvement.

### Methodological finding for the record

A row-count-based support filter and a market-diversity-based support filter both failed to isolate the failure mode. The mechanism is within-market share concentration — high variance in fitted per-market share residuals — and requires variance-sensitive regularization to target. This finding generalizes to any airline logit model calibrated on a heterogeneous market population that includes ULCCs or leisure-focused carriers.

### COST-BASIS: SKYWEST CPA MARKUP

Delta mainline does not operate the E175 aircraft used on the AUS-SLC route. SkyWest Airlines (carrier code OO) operates it under a capacity-purchase agreement, with the marketed service branded as Delta Connection. This has direct implications for the cost side of Verdict 1.

Delta's Form 41 P-5.2 filing shows no aircraft type in the 70-82 seat band. When the pipeline searched under `UNIQUE_CARRIER = 'DL'`, no matching E175 record was found. Under `UNIQUE_CARRIER = 'OO'`, aircraft type 673 emerged as SkyWest's dominant E175 (approximately 115K departures in 2025 Q2, mean 73.8 seats per departure).

Delta pays SkyWest at approximately SkyWest's operating cost plus a documented capacity-purchase markup — industry-typical range 8-15%, with 12% used as central estimate. This adjustment raises effective CASM from SkyWest's raw 6.566 c/ASM to 7.354 c/ASM. Applied to Verdict 1: contribution shrinks from \$9.10M to \$8.61M. All verdicts hold GO under the more conservative cost basis.

### VERDICT 1 RULE RECONCILIATION

An earlier iteration used a load-factor-based rule for Verdict 1: GO iff `expected_LF ≥ breakeven_LF`. Layer 3's introduction of a contribution-based rule created an apparent inconsistency: on ATL-SAT under the shrunk model, Verdict 1 (local-only) came back NO\_GO under LF at 20.1% expected, 34.4% breakeven, while Verdicts 2 and 3 (local + feed) came back GO under contribution.

A retroactive normalization pass reran Verdict 1 with the contribution rule on all three featured markets. Findings: both rules agree GO on AUS-SLC. Both rules agree NO\_GO on ATL-SAT's local-only economics under the shrunk model — the plane genuinely doesn't fill up on Delta's local point-to-point demand alone, even under the corrected share estimate. Local revenue \$21.98M does not cover local cost \$37.65M.

The apparent inconsistency was scope-level, not rule-level: Verdict 1 evaluates local-only viability; Verdicts 2 and 3 evaluate local + feed viability. Both are correct within their scope. Documentation was corrected accordingly, and the pipeline was normalized to the contribution rule throughout.

### ATTRIBUTION LEVERAGE DOES NOT MONOTONICALLY TRACK FEED SHARE

The initial hypothesis was that attribution leverage would scale with feed share of revenue — high-feed markets would show high leverage; low-feed markets would show low leverage.

The three-market rigorous comparison contradicts this:

| Market  | Feed share | Leverage |
|---------|------------|----------|
| AUS-SLC | 13%        | 5.7%     |
| ATL-SAT | 54%        | 33.3%    |
| MIA-SEA | 9%         | 1.3%     |

MIA-SEA has lower feed share than AUS-SLC (9% vs. 13%) yet lower leverage (1.3% vs. 5.7%). Feed share is a necessary but not sufficient condition for attribution leverage. The causal drivers include not only feed share but the relative per-mile yields of the trunk versus its feeders, the standalone-fare availability for feed endpoints, and the direction and magnitude of the Shapley-vs-mileage delta per feed itinerary.

This is a substantive research finding that emerged only because the three-market comparison spanned two feed shares below 15% (AUS-SLC and MIA-SEA) rather than only one.

### DIRECTION OF THE SHAPLEY-VS-MILEAGE DELTA

On AUS-SLC, Shapley credits the AUS-SLC segment less than mileage does — by an average of \$79 per feed itinerary. The economic explanation: short regional segments from SLC (SLC-Boise, SLC-Jackson Hole, SLC-Kalispell) carry disproportionately high fares per mile due to regional monopoly pricing. Distance-based mileage proration over-credits the AUS-SLC trunk relative to the fares those segments could earn independently; Shapley’s value-weighted split corrects this.

The direction depends on the relative per-mile yields of trunk versus feeders. Markets where the trunk has lower per-mile yield than the feeders (like AUS-SLC) will show negative Shapley-minus-mileage on the trunk. Markets where the trunk has higher per-mile yield will show the reverse. This is not a universal-direction mechanism.

### POPULATION-SCALE SCREENER AND ITS ERROR BAR

The scalable variant of the pipeline (aggregate-based on-demand engine) was applied to all 921 US spoke-to-hub markets in the Wave 1 candidate set. Full sensitivity grids (48 combinations  $\times$  2 regimes = 96 grid points per market) were precomputed in approximately six minutes.

*Screener finding.* 106 of 921 markets (11.5%) contained verdict flips somewhere in their pre-registered sensitivity space. The flipping-market subpopulation skewed slightly smaller (median 1,400 local sample passengers vs. 1,827 for non-flipping). Ten markets flipped on all 48 grid points — the mileage-vs-Shapley disagreement is deterministic for these regardless of parameter choice.

*MIA-SEA validation.* The unique market meeting joint criteria of AUS-SLC-comparable size (2,000-6,000 local sample) + verdict-flipping + Delta dominant was MIA-SEA (screener leverage 36%). Rigorous re-analysis under the full pipeline showed 1.3% leverage — no flip. This is not a bug in either pipeline; it is the honest correspondence between fast aggregate approximation and full rigorous analysis. The screener’s 106-flip finding therefore represents an upper bound of potentially-flipping markets.

*Approximation source.* The on-demand engine reconstructs features like `log_fare` as  $\log(\sum \text{fare} \times \text{pax} / \sum \text{pax})$  rather than as passenger-weighted means, because the aggregate parquet stores passenger-weighted sums rather than per-itinerary means. These two forms differ mathematically whenever the underlying quantity correlates with passenger volume within a group.

### ITINID VS MKTID AS JOIN KEY

DB1BMarket’s `ItinID` is not a unique row key — it identifies a roundtrip, and a single roundtrip appears as two directional rows sharing one `ItinID` but with distinct `MktIDs`. An early Layer 3 build used `ItinID` for a merge

between the attribution table and the feed economics table, which would silently produce a cross-join and double-count. The correct per-directional-row key is `MktID`. A defensive `is_unique` check on the merge key was added.

#### **STANDALONE-FARE SPARSITY ON THIN-SPOKE MARKETS**

On ATL-SAT, 47.8% of feed itineraries have no standalone fare data because San Antonio lacks nonstop service to most cities passengers connect to via Atlanta. Shapley collapses to mileage-equivalent for those itineraries. This is a structural feature of thin-spoke markets, not a bug. Full Shapley leverage on such markets requires a  $v(B)$  estimation model or an alternate data source.

#### **NEGATIVE SHAPLEY VALUES**

The Shapley two-player formula can produce negative allocations to one segment when the standalone fare on the other segment plus half the coalition-gain term exceeds the full coalition fare. This occurs when the connecting fare is unusually low relative to the sum of standalone fares (distressed inventory, promotional pricing, etc.).

DESTER reports negative  $\phi_A$  values honestly rather than flooring at zero — 115 of 2,572 AUS-SLC feed itineraries (approximately 4.5%) had negative Shapley allocation, and this is recorded in the output as `phi_A_negative` per itinerary. Zero-flooring would hide a real economic signal.